Centralized SIEM System Using Splunk Enterprise for Real-Time Multi-Platform Log Aggregation and Threat Detection

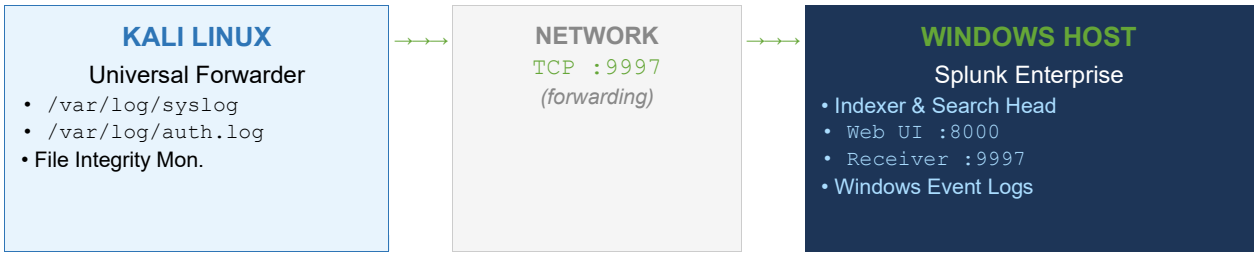
A complete, hands-on lab manual to deploy a production-grade Splunk monitoring environment from scratch, covering installation, configuration, data forwarding, log analysis, and dashboard creation.

Field	Value
Document Type	Technical Implementation Guide
Splunk Version	Splunk Enterprise
Host OS	Windows Server
Forwarder OS	Kali Linux
Data Port	TCP 9997
Web Interface Port	TCP 8000
Monitoring Scope	Windows Events, Linux Syslog, File Integrity

Overview & Architecture

This guide walks you through building a complete Splunk monitoring ecosystem. You will deploy Splunk Enterprise on a Windows host machine that acts as the central indexer and search head, then install a Universal Forwarder on a remote Kali Linux machine that ships logs over the network to port `9997`. A custom dashboard will surface real-time KPIs and remote syslog activity once the pipeline is live.

Architecture Diagram



Prerequisites

- Two machines on the same network (physical, VM, or VirtualBox host-only adapter)
- Windows 10/11 or Windows Server — minimum 8 GB RAM, 20 GB free disk
- Kali Linux (or any Debian-based distro) — minimum 4 GB RAM
- Administrator/root access on both machines
- Internet access for downloading installers
- Firewall rules allowing TCP 9997 and TCP 8000 between machines

NOTE	Throughout this guide, replace 192.168.x.x with your actual IP addresses. Run ipconfig (Windows) or ip addr (Linux) to find them.

Part 1 — Installing Splunk Enterprise on Windows

1 Download Splunk Enterprise

1. Open a browser and navigate to: https://www.splunk.com/en_us/download/splunk-enterprise.html
2. Create a free Splunk account or sign in if you already have one.
3. Select Windows as the platform and download the .msi installer (64-bit).

2 Run the Installer

4. Right-click the downloaded .msi file and select Run as administrator.
5. Accept the License Agreement.
6. Choose Custom install and accept the default path:

```
C:\Program Files\Splunk
```

7. On the Credentials screen, set the administrator username and a strong password — save these in a password manager.
8. Leave Start Splunk Enterprise at boot checked.
9. Click Install and wait for completion (2–5 minutes).

3 Verify Splunk is Running

Open a PowerShell window as Administrator and run:

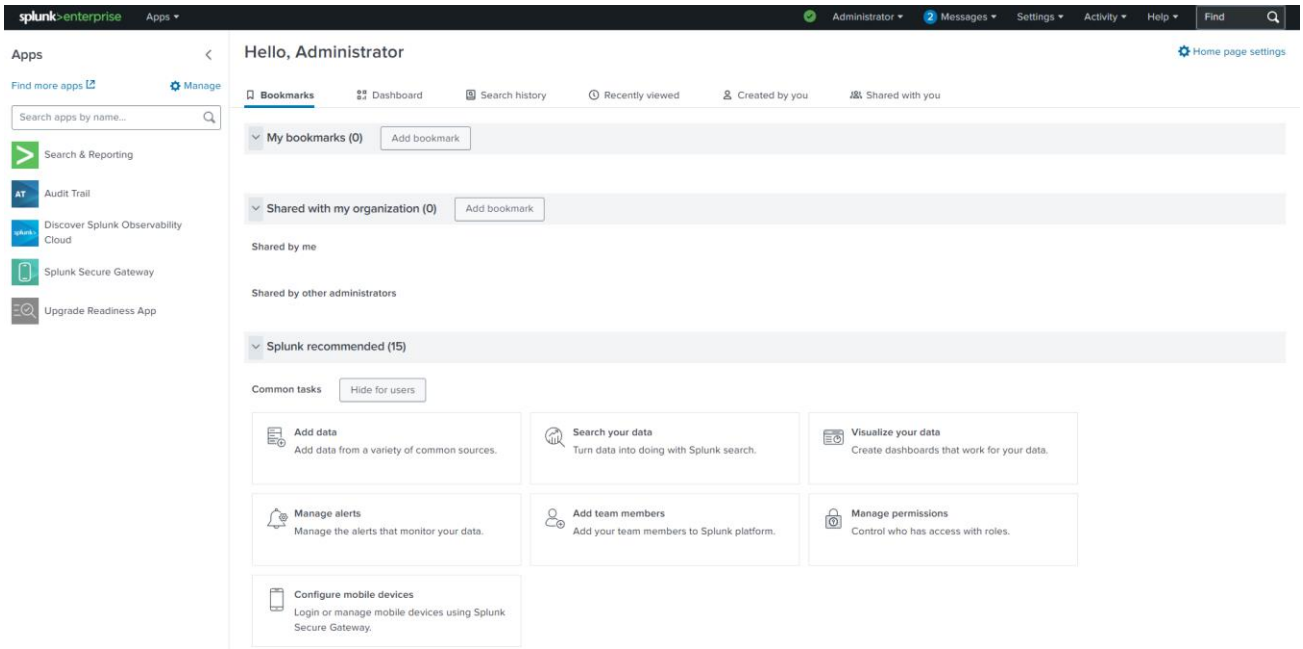
```
# Check Splunk service status
Get-Service -Name Splunkd

# Start if not running
Start-Service -Name Splunkd
```

Expected output:

```
Status   Name               DisplayName
------   ----               -----------
Running  Splunkd            Splunk Service
```

4 Access the Web Interface

10. Open your browser and go to: `http://localhost:8000`
11. Log in with the administrator credentials you set during installation.
12. You should see the Splunk Enterprise home screen.



Part 2 — Configuring Windows Data Inputs

2.1 Windows Event Logs

Windows Event Logs capture system, application, and security events. Splunk can index these natively without additional agents.

13. In the Splunk web UI, navigate to Settings > Data Inputs.
14. Click Local Event Log Collection.
15. Click New Local Event Log Collection.
16. Select the following event log channels:
    - Application
    - Security
    - System
17. Set the index to main (or create a dedicated windows_events index).
18. Click Next and then Review & Finish.

Part 3 — Installing the Universal Forwarder on Kali Linux

1 Download the Universal Forwarder

On the Kali Linux machine, open a terminal and run:

```
# Check system architecture first
uname -m    # Should output: x86_64

# Download Splunk Universal Forwarder (64-bit .deb package)
# Replace X.X.X and <build> with the current version numbers from splunk.com
wget -O splunkforwarder.deb \

'https://download.splunk.com/products/universalforwarder/releases/9.x.x/linux/splunkfo
rwarder-x.x.x-<build>-linux-2.6-amd64.deb'

# Verify download completed
ls -lh splunkforwarder.deb
```

2 Install the Forwarder Package

```
# Install the .deb package
sudo dpkg -i splunkforwarder.deb

# If dependencies are missing, fix with:
sudo apt-get install -f -y

# Verify installation path
ls /opt/splunkforwarder/
```

3 Start the Forwarder & Accept the Licence

```
# Start Splunk and accept the licence agreement
sudo /opt/splunkforwarder/bin/splunk start --accept-license

# Set admin credentials when prompted
# Username: admin
# Password: (choose a strong password)

# Enable start-on-boot
sudo /opt/splunkforwarder/bin/splunk enable boot-start
```

4 Point the Forwarder at the Windows Indexer

This tells the forwarder to ship data to your Windows host:

```
# Replace 192.168.1.100 with your actual Windows host IP address
sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.1.100:9997 \
  -auth admin:YourForwarderPassword

# Verify the forward-server was added
sudo /opt/splunkforwarder/bin/splunk list forward-server
```

Expected output:

```
Active forwards:
  192.168.1.100:9997
Configured but inactive forwards:
  (none)
```

Part 4 — Configuring Linux Data Inputs

4.1 Syslog File Monitoring

Configure the Universal Forwarder to tail the core Linux syslog files and forward them in real time.

Create the inputs.conf file

```
# Create or edit the inputs configuration
sudo nano /opt/splunkforwarder/etc/system/local/inputs.conf
```

Add the following content:

```
[monitor:///var/log/syslog]
disabled = false
index = linux_logs
sourcetype = syslog
followTail = 0
ignoreOlderThan = 7d

[monitor:///var/log/auth.log]
disabled = false
index = linux_logs
sourcetype = linux_auth

[monitor:///var/log/kern.log]
disabled = false
index = linux_logs
sourcetype = linux_kern

[monitor:///var/log/dpkg.log]
disabled = false
index = linux_logs
sourcetype = dpkg
```

4.2 File Integrity Monitoring (FIM)

Monitor critical system files for unexpected changes — a core security control.

Configure FIM inputs

Add these sections to `/opt/splunkforwarder/etc/system/local/inputs.conf`:

```
# File Integrity Monitoring — system binaries
[batch:///etc/passwd]
move_policy = sinkhole
sourcetype = fim
```

```
index = linux_fim
_TCP_ROUTING = *

[batch:///etc/shadow]
move_policy = sinkhole
sourcetype = fim
index = linux_fim

[batch:///etc/sudoers]
move_policy = sinkhole
sourcetype = fim
index = linux_fim

# Monitor /etc directory for new/changed files
[monitor:///etc]
disabled = false
recursive = true
index = linux_fim
sourcetype = fim_etc

# Monitor cron directories
[monitor:///var/spool/cron]
disabled = false
index = linux_fim
sourcetype = fim_cron
```

4.3 Configure the Outputs (props & transforms)

Create outputs.conf

```
sudo nano /opt/splunkforwarder/etc/system/local/outputs.conf
```

Content:

```
[tcpout]
defaultGroup = splunk_indexer
maxQueueSize = 7MB
useACK = false

[tcpout:splunk_indexer]
server = 192.168.1.100:9997
compressed = true
sendCookedData = true
```

4.4 Restart and Verify the Forwarder

```
# Restart to pick up new configuration
sudo /opt/splunkforwarder/bin/splunk restart
```

```
# Tail the forwarder log to confirm data is being sent
sudo tail -f /opt/splunkforwarder/var/log/splunk/splunkd.log | grep -i forward

# You should see lines like:
# INFO  TcpOutputProc - Connected to 192.168.1.100:9997
```

Part 5 — Creating Indexes on Splunk Enterprise

Indexes organize your data by type, control retention, and allow efficient searching. Create dedicated indexes before data starts flowing so logs are not placed in the default 'main' index.

Create Indexes via the UI

1. Go to Settings > Indexes > New Index.
2. Create the following indexes:

Index Name	Purpose
linux_logs	Syslog, auth.log, kern.log from Kali Linux forwarder
linux_fim	File Integrity Monitoring events from Kali Linux
perfmon	Windows Performance Counter metrics
windows_events	Windows Event Log (optional, or use 'main')

3. For each index, set Max Size (GB) to a suitable limit (e.g. 5 GB for a lab).
4. Leave Frozen Path blank for now (data will be deleted when max size is reached).
5. Click Save after each index.

Part 6 — Building the Monitoring Dashboard

This dashboard provides a single-pane view of system health KPIs and remote syslog activity. It combines Windows performance metrics with Linux log data from the forwarder.

6.1 Create a New Dashboard

6. In the Splunk web UI, click Dashboards in the left nav (or Apps > Search & Reporting > Dashboards).
7. Click Create New Dashboard.
8. Name it: **SOC Home — System Monitoring**
9. Select Classic Dashboard (not Dashboard Studio for broadest compatibility).
10. Set permissions to Shared in App so all users can view it.
11. Click Create Dashboard.

6.2 Add Dashboard Panels

Click Edit > Add Panel for each panel below. Use Add Search and paste the SPL query provided.

Panel 1 — Events per Hour (Remote Syslog)

Shows the volume of Linux syslog events arriving from the forwarder over time.

```
index=linux_logs sourcetype=syslog
| timechart span=1h count as "Syslog Events"
| rename _time as "Hour"
```

- Visualisation type: Line Chart
- Time range: Last 24 Hours
- Panel title: Remote Syslog — Events per Hour

Panel 2 — Top Syslog Sources

```
index=linux_logs sourcetype=syslog earliest=-24h
| rex field=_raw "(?P<prog>[a-zA-Z0-9_-]+)\["
| stats count by prog, host
| sort -count
| head 10
| rename prog as "Process", host as "Host", count as "Event Count"
```

- Visualisation type: Bar Chart
- Panel title: Top Syslog Processes

Panel 3 — Failed SSH Logins (Security KPI)

```
index=linux_logs sourcetype=linux_auth
| search "Failed password" OR "authentication failure"
| rex "from (?P<src_ip>\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})"
| stats count as "Failed Logins" by src_ip, host
| sort -"Failed Logins"
| head 10
```

- Visualisation type: Statistics Table
- Panel title: Failed SSH Login Attempts — Top Sources

Panel 4 — Windows CPU Utilisation

```
index=perfmon object="Processor" counter="% Processor Time" instance="_Total"
| timechart span=5m avg(Value) as "CPU %"
```

- Visualisation type: Line Chart
- Y-axis max: 100
- Panel title: Windows Host — CPU Utilisation (%)

Panel 5 — Windows Memory Available

```
index=perfmon object="Memory" counter="Available MBytes"
| timechart span=5m avg(Value) as "Available MB"
```

- Visualisation type: Area Chart
- Panel title: Windows Host — Available Memory (MB)

Panel 6 — Windows Security Events (Single Value)

```
index=main OR index=windows_events sourcetype=WinEventLog:Security earliest=-1h
| stats count as "Security Events"
```

- Visualisation type: Single Value
- Panel title: Security Events — Last Hour

Panel 7 — File Integrity Alerts

```
index=linux_fim earliest=-24h
| stats count by source, host
| sort -count
| rename source as "Monitored Path", host as "Host", count as "Access Count"
```

- Visualisation type: Statistics Table
- Panel title: File Integrity — Monitored Path Activity

Panel 8 — Network Traffic (Performance Counter)

```
index=perfmon object="Network Interface" counter="Bytes Total/sec"
| timechart span=1m avg(Value) as "Bytes/sec"
```

- Visualisation type: Line Chart
- Panel title: Windows Host — Network Throughput

6.3 Arrange and Save the Dashboard

12. Click and drag panels to arrange them in a 2-column or 3-column grid.
13. Recommended layout: Row 1 — Syslog Events over Time (full width). Row 2 — CPU, Memory, Network (three equal columns). Row 3 — Security Events single value, Failed SSH table. Row 4 — FIM table, Top Syslog processes.
14. Click Save to commit the layout.
15. Click Set as Home Dashboard so it loads by default on login.

Part 7 — Testing & Verification

7.1 End-to-End Connectivity Check

Test	Command & Expected Result
Splunk running on Windows	Get-Service Splunkd → Running
Forwarder running on Kali	sudo /opt/splunkforwarder/bin/splunk status → splunkd is running
Port 9997 open on Windows	nc -zv <WINDOWS_IP> 9997 → succeeded (run from Kali)
Port 8000 accessible	curl -I http://<WINDOWS_IP>:8000 → HTTP 200 or redirect
Forwarder connected	splunkd.log grep 'Connected to' → active connection entry

7.2 Verify Data Ingestion in Splunk Search

In the Splunk web UI, open the Search & Reporting app and run these queries:

Check Linux syslog is arriving

```
index=linux_logs earliest=-10m
| stats count by host, sourcetype
```

Check Windows Event Logs

```
index=main sourcetype=WinEventLog earliest=-10m
| stats count by EventCode | sort -count | head 10
```

Check Performance Counter data

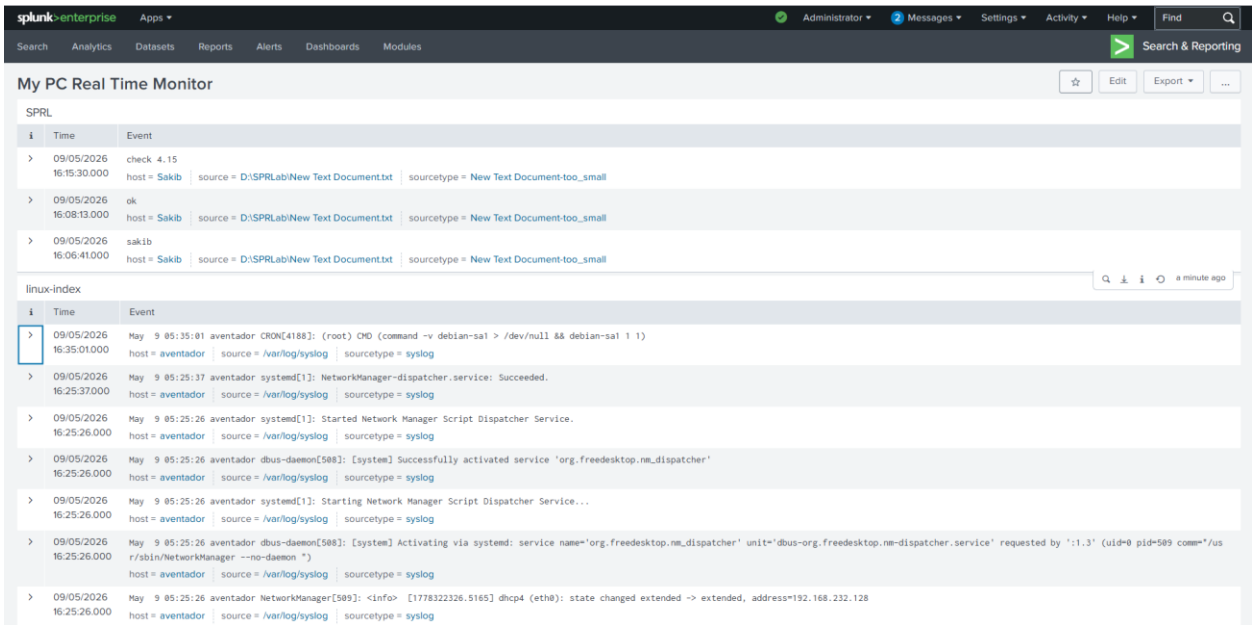
```
index=perfmon earliest=-10m
| stats count by object, counter
```

Generate a test syslog event from Kali

```
# On the Kali Linux machine, generate a test syslog entry
logger -p local0.info 'SPLUNK_TEST: This is a verification message from Kali Linux'

# Then search for it in Splunk
# index=linux_logs SPLUNK_TEST
```

7.3 Simulate Security Events for Dashboard Testing

```
# On Kali Linux — generate multiple failed SSH attempts to yourself
# (This creates real auth.log entries for the dashboard)
for i in {1..5}; do
  ssh -o StrictHostKeyChecking=no baduser@127.0.0.1 2>/dev/null || true
  sleep 1
done

# Verify entries appeared in auth.log
sudo grep 'Failed' /var/log/auth.log | tail -5
```

NOTE Simulated failures against localhost are safe and will not affect your system. They simply populate auth.log so you can verify the Splunk dashboard panel is working end-to-end.

7.4 Dashboard